Lecture no 4:

- Distributed DW → • economics & technology greatly favour a single centralized DW

  - Types → • geographically distributed (information needed both locally & globally)

    • technologically distributed DW (logically one DW but data is store on multiple stores)

    Advantage → • entry cost is cheap - large centralized hardware is expensive.

  - As DW starts to expand network data communication starts playing an important role.

- Data modeling / DB design - Basics
  - Process of creating data model by analyzing requirements needed to support an organization.
  - Sometimes called as database modeling / design b/c data model is implemented in database.
- Data models (provide definition & format of data)
  - graphical representation of data within specific area of interest.
    - Enterprise data model → represents integrated data requirements of complete organization
    - Subject Area data model → represent data requirement of single buisness area.

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- Conceptual design (e.g. ER-diagram, UML)

Conceptual
↓
logical
↓
physical.

  - Transforms data requirements to conceptual model.
  - Conceptual model → describes data entities, relationships, constraint on high level.

- Logical design (e.g., relational model, dimensional model) (e.g., Tables, Columns)
  - Maps conceptual data model to logical data model used by DBMS.

- Physical design (e.g., Tablespaces, indexes)
  - Creates internal structure needed to efficiently store data.

① Conceptual model:-

- Highest conceptual grouping of ideas
- least amount of detail
- Conceptual design is usually done using Unified Modeling Language (UML) → • class diagram, component diagram, object diagram
  - For data modeling only class diagram are used.

② logical model:-

- logical design arranges data into logical structure.
  e.g., In RDB, storage objects are tables which store data in rows & columns.

③ Physical model:-

- physical design specifies physical configuration of database on storage media.

- Data model in DW → • Manage complex data relationship.
  • Transformation & integration from various system need to be maintained.
  • Provide means of supplying user with a roadmap through data & relationship.

- Conceptual model in DW
  • Goal (purpose, question of interest)
  • Subject (who bought products, who sold, when sold, what sold)
  • For conceptual design in DW techniques e.g. like E/R or UML are not appropriate.
  • E/R are (• remove redundancy
    • facilitate retrieval of individual records)
  • Components (facts, measures, dimensions, Hierarchies)

Examples → • Multidimensional Entity relationship (ME/R) Model
  • Multidimensional UML (m UML)

mUML.

- ME/R Model → • It creates intuitive representation of multidimensional data (high performance access)

- Specialization → • Introduced specialized elements on existing ER model without reducing its flexibility.
  • new elements represent multidimensional data but easy to learn.

- Representation of multidimensional semantics → efficient representation of complex, multidimensional relationships.
  • Fact node (represents primary fact)
  • Level node (represent classification levels.)
  • Special Binary classification edge (supports hierarchical relationships)